

Syed Munawar Saeed Kazmi

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PROFILE

Robotics Systems Engineer with an MSc (Commendation) in AI and Robotics, specialising in humanoid HRI, sim-to-real transfer, and edge AI deployment. Led HRI research on the iCub and JD humanoid platforms, achieving a 30% measurable trust improvement (RoSAS) validated across 200+ trials, with a co-authored manuscript under review at MDPI Robotics. Delivers deterministic, safety-critical systems from simulation to deployed hardware, and ships production software end to end: most recently a complete institutional management platform, built solo and running live.

TECHNICAL SKILLS

Robotics & Software: C++20, Python, ROS2 (Nav2, ROS Control, Plugin Development), Git, CI/CD, Docker

Simulation & Digital Twins: Isaac Sim, MuJoCo, Gazebo, Unity, Sim-to-Real Transfer

AI & Perception: PyTorch, TensorFlow, TensorRT, Computer Vision, Imitation Learning, VLA Models, LLM Integration

Hardware & Embedded: iCub Humanoid, Unitree Go1, Jetson Nano, ESP32, LoRa Mesh, Bare-Metal C/C++

Full-Stack & Production: React, TypeScript, PostgreSQL (row-level security), Supabase, REST, Netlify

FEATURED RESEARCH

iCub & JD Humanoid Sim-to-Real Pipeline for Trust-Calibrated HRI — MSc Thesis

University of Hertfordshire

Isaac Sim, Digital Twin, ROS2, Physical Deployment | Supervised by Prof. Patrick Holthaus

- Led a full-stack sim-to-real pipeline across two humanoid platforms, building a high-fidelity Isaac Sim digital twin and transferring hybrid verbal and non-verbal intent communication strategies to physical hardware.
- Achieved a 30% improvement in human trust scores (RoSAS) and 98% intent clarity across 200+ trials through quantitative user studies; co-authored manuscript submitted to MDPI Robotics, under review.

EXPERIENCE

Full-Stack Software Engineer (Contract) · Safina International

Multan, Pakistan · Jul 2026 – Present

React, TypeScript, Supabase | github.com/munawarkazmi/safina-portal-showcase

- Sole engineer of the institute's complete management platform, live at portal.safinainternational.com: seven user roles, prorated billing engine, payroll, library, facility bookings, digital ID cards, and a database-trigger audit trail.
- Enforced security entirely through PostgreSQL row-level security and guarded functions; financial records are append-only, corrected by recorded reversals rather than deletion.

Visiting Lecturer, AI and Robotics (Contract) · Pak-Turk Maarif Schools

Multan, Pakistan · Aug 2025 – Present

- Designed an ML curriculum for 30+ students; teams achieved 92% model accuracy with 28% average error reduction on custom classification tasks.

Embedded Systems Engineer (Contract) · Muxtronics

Multan, Pakistan · Jan 2025 – May 2025

- Engineered deterministic C++ firmware for real-time image processing on ESP32, reducing peak RAM usage by 60%; architected a LoRa mesh protocol achieving sub-3-second alert latency across 100 m field trials in a deployed safety-critical product.

Robotics Integration Specialist · Robot House, University of Hertfordshire

Hatfield, UK · Jan 2024 – May 2024

- Architected a gesture recognition and motion planning pipeline in Unity and C++ for the iCub humanoid, reducing user frustration by 40% and improving HRI task efficiency by 25%.
- Deployed PyTorch-based non-verbal communication models onto iCub's real-time control loop, lifting user engagement from 4.7 to 5.2 (7-point Likert) across structured interaction trials.

Research Associate · University of Hertfordshire

Hatfield, UK · May 2023 – Dec 2023

- Fine-tuned TensorFlow LLMs for real-time robotics inference on the Unitree Go1 quadruped, cutting inference latency by 29% across 200 trials.
- Integrated optimised models into the control loop, improving platform responsiveness by 35% with 99.5% runtime stability.

KEY PROJECTS

ROS2 Autonomous Navigation and Deterministic Planning github.com/munawarkazmi/ros2-dynamic-path-planning

- C++20 Global Planner plugin for ROS2 Nav2 (A* and D* Lite): 22% faster replanning and 24% travel time reduction with strict protective-stop safety guarantees, validated over 200 reproducible trials in dynamic environments.

ROS2 LLM Safety Verifier github.com/munawarkazmi/ros2-llm-safety-verifier

- Real-time safety layer intercepting LLM-hallucinated trajectories before execution: 94% unsafe-trajectory catch rate at under 50 ms on Jetson Orin Nano; navigation success improved from 57% to 91% across 103 hardware trials on TurtleBot3.

Edge AI Optimisation on Jetson Nano [TensorRT, PyTorch, CNN](#)

- TensorRT optimisation of CNN and gesture recognition models: 40% latency reduction at 96% accuracy, enabling real-time decentralised motor control across a multi-robot system.

EDUCATION

MSc Artificial Intelligence and Robotics, Commendation · University of Hertfordshire

Hatfield, UK · Jan 2023 – Aug 2024

BEng Mechatronics Engineering · National University of Sciences and Technology (NUST)

Islamabad, Pakistan · Sep 2018 – May 2022

PUBLICATIONS

Kazmi, M. (2025). Human-Robot Interaction for Companionship: Verbal and Non-Verbal Communication Cues in Humanoid Robots. *Submitted to MDPI Robotics, under review.*